CREDIT RISK MODELING

& LOAN DEFAULT ANALYSIS

Home Credit Default Risk | 307,511 Applications | XGBoost

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307K	8.06%	0.75
Applications	Default Rate	AUC Score

THE PROBLEM WITH HOW LENDING DECISIONS ARE MADE TODAY

WRONG APPROVALS

High-risk applicants slip through manual underwriting — default, generating direct financial losses the business cannot recover.

WRONG REJECTIONS

Creditworthy applicants are turned away because their profiles don't fit traditional scoring models — destroying revenue and market share.

Can data price risk better than intuition — consistently, at scale, and in real time?

THE PORTFOLIO AT A GLANCE

307,511

Loan Applications

Home Credit dataset

8.06%

Defaulted

24,825 applicants

91.94%

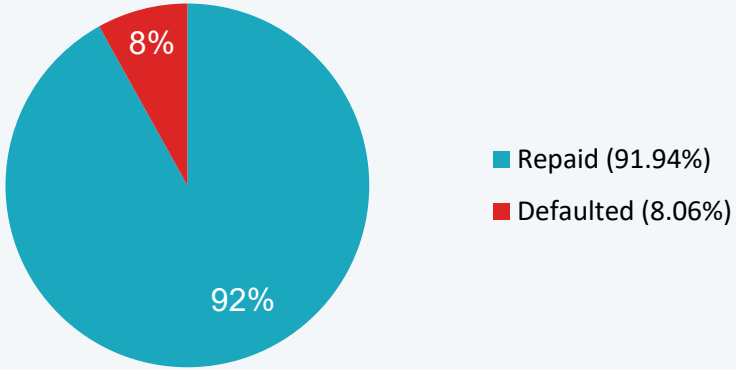
Repaid

282,686 applicants

~11:1

Class Imbalance

Repaid vs Defaulted

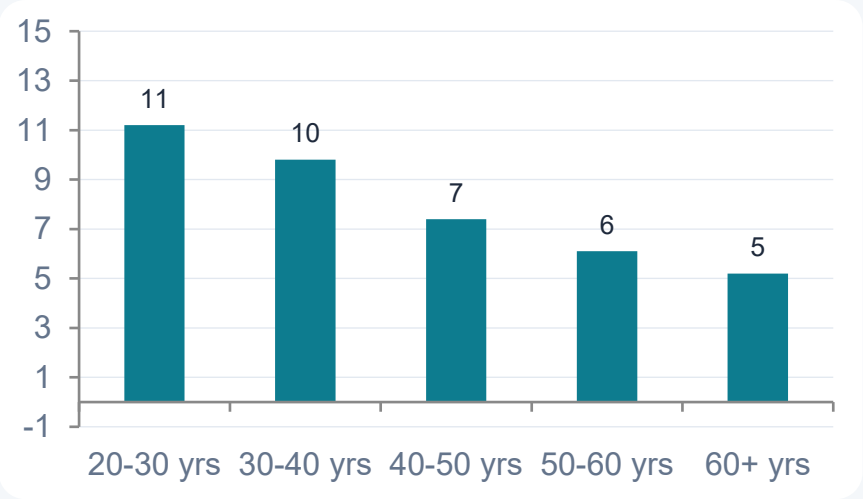


⚠ THE STRUCTURAL BLIND SPOT

An 11:1 imbalance means a model that approves everyone achieves 92% accuracy — while missing every single defaulter. Accuracy is the wrong metric. The real question is: who are the 8%?

WHO ACTUALLY DEFAULTS?

DEFAULT RATE BY AGE GROUP



10.14%

Male applicants

vs females at 7.00%

12.31%

Renters

vs house/apt owners at 7.80%

11.70%

Living with parents

vs co-op apartment at 7.93%

9.94%

Civil marriage

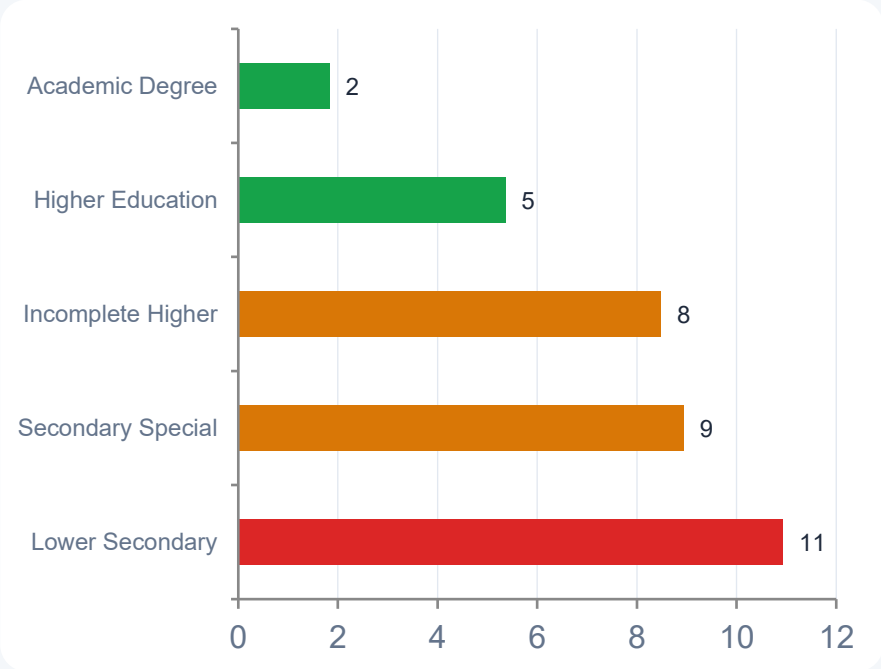
vs married at 7.56%

Default risk has a demographic fingerprint — younger, male, asset-light applicants default at structurally higher rates

OCCUPATION & EDUCATION ARE THE STRONGEST BEHAVIOURAL SIGNALS

DEFAULT RATE BY OCCUPATION

DEFAULT RATE BY EDUCATION



What you do and how far you studied predicts default better than how much you earn

THE COUNTERINTUITIVE FINDING: INCOME DOESN'T SEPARATE RISK

MEDIAN VALUES — DEFAULTERS vs NON-DEFAULTERS

Variable	Non-Defaulter	Defaulter	Separates Risk?
Total Income	$ 147,150	$ 135,000	❌ Barely
Credit Amount	$ 513,000	$ 544,500	❌ Barely
Annuity Amount	$ 25,155	$ 26,018	❌ Barely
Credit/Income Ratio	3.2×	4.6×	✓ Yes — ratio matters
Annuity/Income Ratio	0.17	0.22	✓ Yes — burden matters

THE RATIO INSIGHT

A $500K loan means something entirely different to someone earning $200K versus $2M.

Raw amounts are noise.
Ratios are the signal.

6 Engineered Features → credit_income_ratio | annuity_income_ratio | credit_goods_ratio | income_per_person | ext_source_mean | ext_source_min

credit_goods_ratio — an engineered feature — ranked in the top 5 SHAP predictors of the final model

EXTERNAL CREDIT SCORES: THE SINGLE STRONGEST PREDICTOR

EXT_SOURCE_1

Correlation with Default: -0.10

56.38% missing

EXT_SOURCE_2

Correlation with Default: -0.16

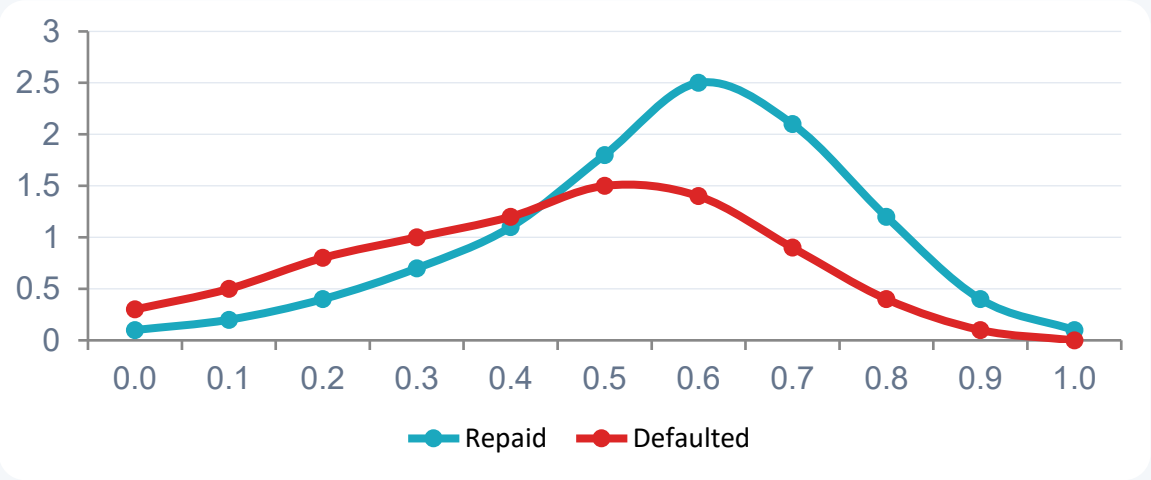
0.21% missing

EXT_SOURCE_3

Correlation with Default: -0.16

19.83% missing

SCORE DISTRIBUTION: REPAID vs DEFAULTED (EXT_SOURCE_2)



⚠ CRITICAL DATA GAP

EXT_SOURCE_1 — the strongest predictor — was missing for 56.38% of all applicants at ingestion.

Every missing value is a lost opportunity to price risk accurately.

Fixing the data pipeline would directly improve model discrimination with zero architectural change.

THE MODEL: WHAT IT LEARNED AND HOW WELL

CHAMPION MODEL — XGBoost | Test Set Performance

0.7536

AUC Score

Strong discriminative power across all thresholds

39.92%

Recall

2 in every 5 actual defaulters correctly identified

0.2998

F1 Score

Best precision-recall balance of all 4 models

-0.004

AUC Drift

From validation to test — zero overfitting

ALL MODELS — VALIDATION SET COMPARISON

Model	AUC	Recall	F1	KS Stat
XGBoost ★	**0.7495**	**0.3955**	**0.2924**	**0.3784**
Logistic Regression	0.7343	0.6541	0.2562	0.3533
Random Forest	0.7099	0.2784	0.2294	0.3096
Decision Tree	0.6635	0.1490	0.1492	0.2645

THE THRESHOLD DECISION: WHY 40% RECALL IS THE RIGHT BUSINESS CHOICE

THRESHOLD TUNING — PRECISION vs RECALL TRADE-OFF (TEST SET)

Threshold	Recall	Defaults Caught	False Alarms	Business Implication
0.30	**70.92%**	3,521	18,597	Review queue overwhelmed — operationally unviable
0.35	62.84%	3,120	14,546	High catch rate but 4× false alarms per true default
0.40 ★	**55.13%**	**2,737**	**11,284**	**Optimal — highest F1, manageable review volume**
0.45	47.63%	2,365	8,503	Conservative — lower cost, more missed defaults
0.50	39.92%	1,982	6,276	Statistical default — misses 60% of defaulters

THE ASYMMETRIC COST OF ERRORS

False Negative

Missed defaulter
→ Full loan lost
Unrecoverable

False Positive

Rejected good applicant
→ Lost interest income
Recoverable opportunity

WHY 0.50 IS NOT A BUSINESS DECISION

0.50 is a statistical default — not a considered choice. At 0.50, the model misses 60% of defaulters. Every miss is an unrecoverable loss.

Threshold 0.40 was selected because it peaks on F1 — the metric that explicitly balances both error types. False alarms are recoverable. Missed defaults are not.

The threshold is a business decision — not a default parameter. It was set where F1 peaks and expected loss is minimised.

THE DATA GAP THAT IS COSTING THE MODEL ITS CEILING

EXT_SOURCE_1

CRITICAL GAP

Correlation with default: -0.10

Missing at ingestion: 56.38%

Applicants imputed: 173,271

→ Zero discriminative power for these records

EXT_SOURCE_2

COMPLETE

Correlation with default: -0.16

Missing at ingestion: 0.21%

Applicants imputed: 646

→ Zero discriminative power for these records

EXT_SOURCE_3

PARTIAL GAP

Correlation with default: -0.16

Missing at ingestion: 19.83%

Applicants imputed: 60,979

→ Zero discriminative power for these records

BUSINESS IMPACT OF THE DATA GAP

173,271

applicants scored with EXT_SOURCE_1 replaced by a constant — 3rd strongest predictor carrying zero signal for these records

56.38%

of the entire portfolio blind to one of the top 3 SHAP predictors — not because the feature is weak, but because the data was never collected

0 changes

to model architecture required — improving the data feed alone directly improves discrimination for 173K+ applicants

Fixing EXT_SOURCE_1 is the single highest-ROI model improvement available — no retraining, no new features, no architecture change

THE RISK TIER FRAMEWORK: ONE MODEL, THREE BUSINESS DECISIONS

LOW RISK

AUTO-APPROVE

% of Portfolio
33.14%

Applicants
101,920

Actual Default Rate
2.20%

Avg Predicted Prob
0.09

Default rate less than 3× below portfolio average. Zero manual underwriting needed.

MEDIUM RISK

MANUAL REVIEW

% of Portfolio
44.06%

Applicants
135,490

Actual Default Rate
6.57%

Avg Predicted Prob
0.26

Largest segment. Model score + SHAP drivers surfaced to underwriter for defensible decision.

HIGH RISK

AUTO-DECLINE

% of Portfolio
22.80%

Applicants
70,105

Actual Default Rate
19.52%

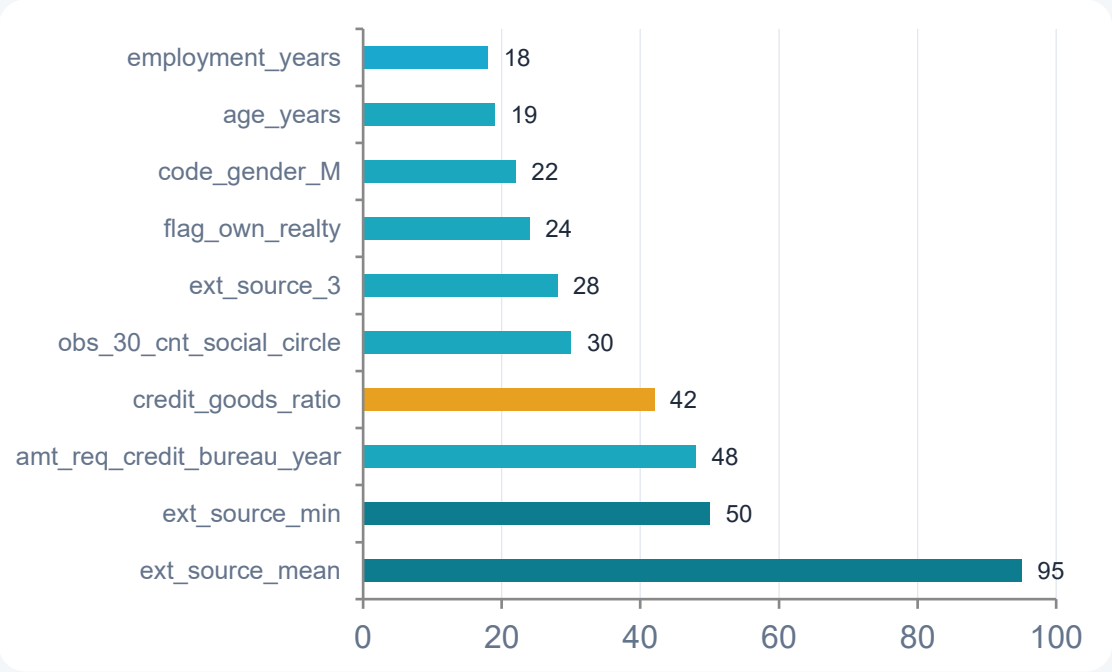
Avg Predicted Prob
0.55

Nearly 1 in 5 defaults. More than double the portfolio average. Automatic decline.

9× default rate gap between Low Risk and High Risk — same model, three entirely different business decisions

WHAT DRIVES EVERY DECISION — MODEL INTERPRETABILITY

TOP SHAP PREDICTORS (by impact magnitude on test set)



TWO APPLICANTS — TWO STORIES

HIGH RISK APPLICANT

Score: 2.893 | Base: 1.746

Driven by: low ext_source_mean (+0.89), low ext_source_min (+0.50), high credit_goods_ratio (+0.32)

LOW RISK APPLICANT

Score: -5.892 | Base: 1.746

All 14 features push score down. ext_source_mean alone contributes -0.95. Zero risk-increasing signals.

Every approval or decline is explainable — no black box, fully defensible to regulators and credit committees

STRATEGIC RECOMMENDATIONS

01

Auto-Approve Low Risk

101,920 applicants | 2.20% default

→ Eliminate manual underwriting cost for 33% of portfolio

02

Human-in-the-Loop for Medium Risk

135,490 applicants | 6.57% default

→ Protect revenue + model SHAP drivers surfaced to underwriters

03

Auto-Decline High Risk

70,105 applicants | 19.52% default

→ Prevent majority of financial losses — threshold at 0.40

04

Occupational Risk Premia

Low-skill laborers 17.15% | Unemployed 36.36%

→ Tighten thresholds for high-default occupations without model retraining

05

Fix EXT_SOURCE Data Pipeline

56.38% of EXT_SOURCE_1 missing at ingestion

→ Directly improve model discrimination — no architecture change needed

THE BOTTOM LINE

"The model prices risk the way a credit analyst would — but at scale and in milliseconds."

Risk Is Not Random

It concentrates in identifiable profiles — the current system is systematically missing them.

Segmentation Works

9× default rate gap end-to-end. One model enables three distinct, defensible business decisions.

The Fix Is Known

Deploy the tier framework, invest in EXT_SOURCE data, revisit threshold in 6 months.